# Intelligent Q & A Chatbot using Multimodal Retrieval- Augmented Generation

Bridging the Gap between static language models and dynamic user provided data

Presented by
Enosh Joseph P A
MAC24MCA-2025

# Introduction

- Chatbots have become an essential tool for information access in applications such as education, customer support, and enterprise systems.
- Most existing chatbot systems rely on pretrained language models with static internal knowledge.
- Such chatbots are unable to effectively answer questions based on user specific documents and often generate inaccurate or hallucinated response.
- Users frequently need to search large documents manually to find relevant information.
- Traditional chatbots cannot reason over uploaded files or multimedia content.

# Existing System

- Many chatbots handle only text.
- Cannot process diagrams, images, or video.
- Generate fluent responses using internal knowledge.
- High risk of hallucination.
- Large language model–based chatbots generate natural and coherent answers based on patterns learned during training, rather than reasoning over user-provided documents or external data sources.
- Lack video-based question answering capability.

# Literature Review

| Paper | Year | Methodology | Description |
|---|---|---|---|
| MDKAG:Retrieval-Augmented Educational QA Powered by a Multimodal Disciplinary Knowledge Graph | 2025 | OCR for documents, text–image extraction from slides, ASR-based transcription for videos, graph-based retrieval | To improve educational question answering by reducing hallucinations and enhancing answer accuracy using multimodal knowledge |
| Multimodal retrieval-augmented generation framework for visually rich knowledge in the architecture domai | 2025 | CLIP, VisualBERT, and Doubao embedding models for multimodal retrieval; Qwen2.5-VL for table parsing; Qwen3 MLLM for multimodal answer generation | Proposes a multimodal RAG framework that unifies text, images, and tables for accurate, traceable knowledge retrieval and generation in architectural design tasks |

| Paper | Year | Methodology | Description |
|---|---|---|---|
| Knowledge graph-extended retrieval augmented generation for question answering | 2025 | KG-based multi-hop retrieval, and similarity-based triple selection to generate explainable answers without model training | training-free question answering framework that integrates Large Language Models (LLMs) with Knowledge Graphs (KGs) to improve accuracy and explainability. |

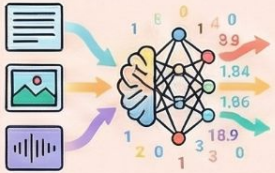
## Proposed System

- The proposed system is an intelligent Question Answering chatbot that responds to user queries based on user-uploaded documents and media.
- The system follows the Retrieval-Augmented Generation (RAG) pipeline, where relevant information is retrieved first and then used to generate accurate answers.
- Unlike traditional chatbots that rely only on internal knowledge, the proposed system grounds responses in retrieved evidence, improving reliability.

- OCR → “text extraction from images and scanned PDFs”
- all-MiniLM → “lightweight and efficient embeddings”
- ChromaDB → “fast vector similarity search”
- LLaMA-3 → “final answer generation with context”

# The Multimodal RAG Chatbot: From Upload to Answer

## The Core Components



### Embedding Model

Converts text from all sources into numerical vectors to capture semantic meaning.

### OCR Model

Extracts text from images and scanned PDFs, making visual content searchable.

### Speech-to-Text Model

Transcribes audio from video files into text, enabling search over video content.

### Vector Database

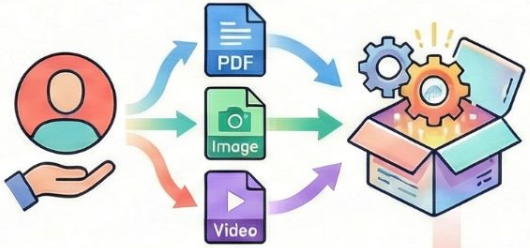
Stores all the numerical vectors (embeddings) for fast and efficient similarity search.

### Large Language Model (LLM)

Generates the final, human-like answer using the user's question and retrieved context.

## The Workflow: Step-by-Step

### 1. Ingestion & Processing




A user uploads a PDF, image, or video, which is processed by parsers, OCR, or speech-to-text models.

### 3. Query & Retrieval

### 4. Generation

### 2. Embedding & Storage

The extracted text is broken into chunks and converted into embeddings stored in a vector database.

### 3. Query & Retrieval

The user's question is also converted to an embedding to find the most relevant context from the database.

### 4. Generation

The retrieved context and the original question are sent to an LLM to generate the final answer.

NotebookLM

# Thank You…